```
All libraries imported successfully!
```

First five rows:

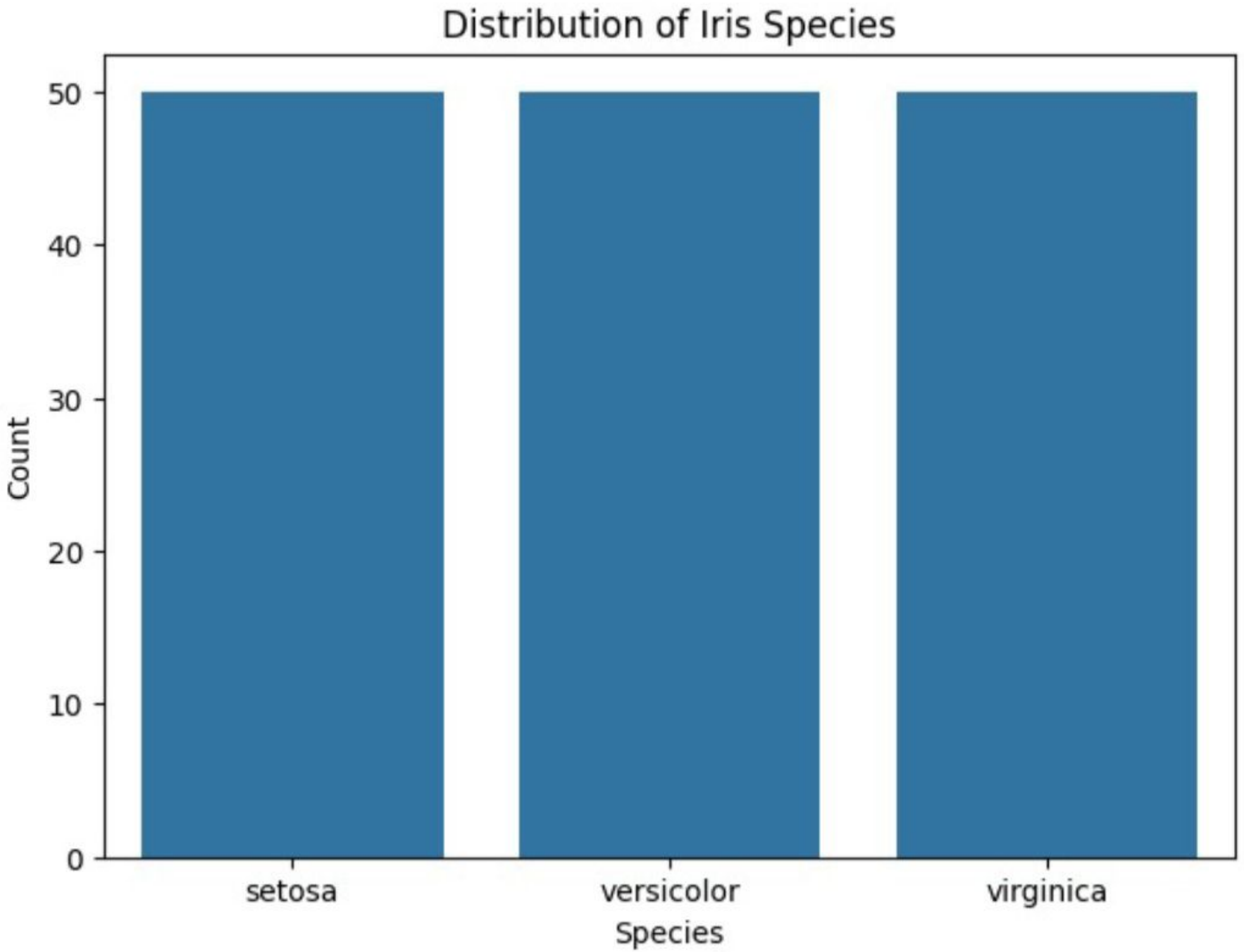
	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

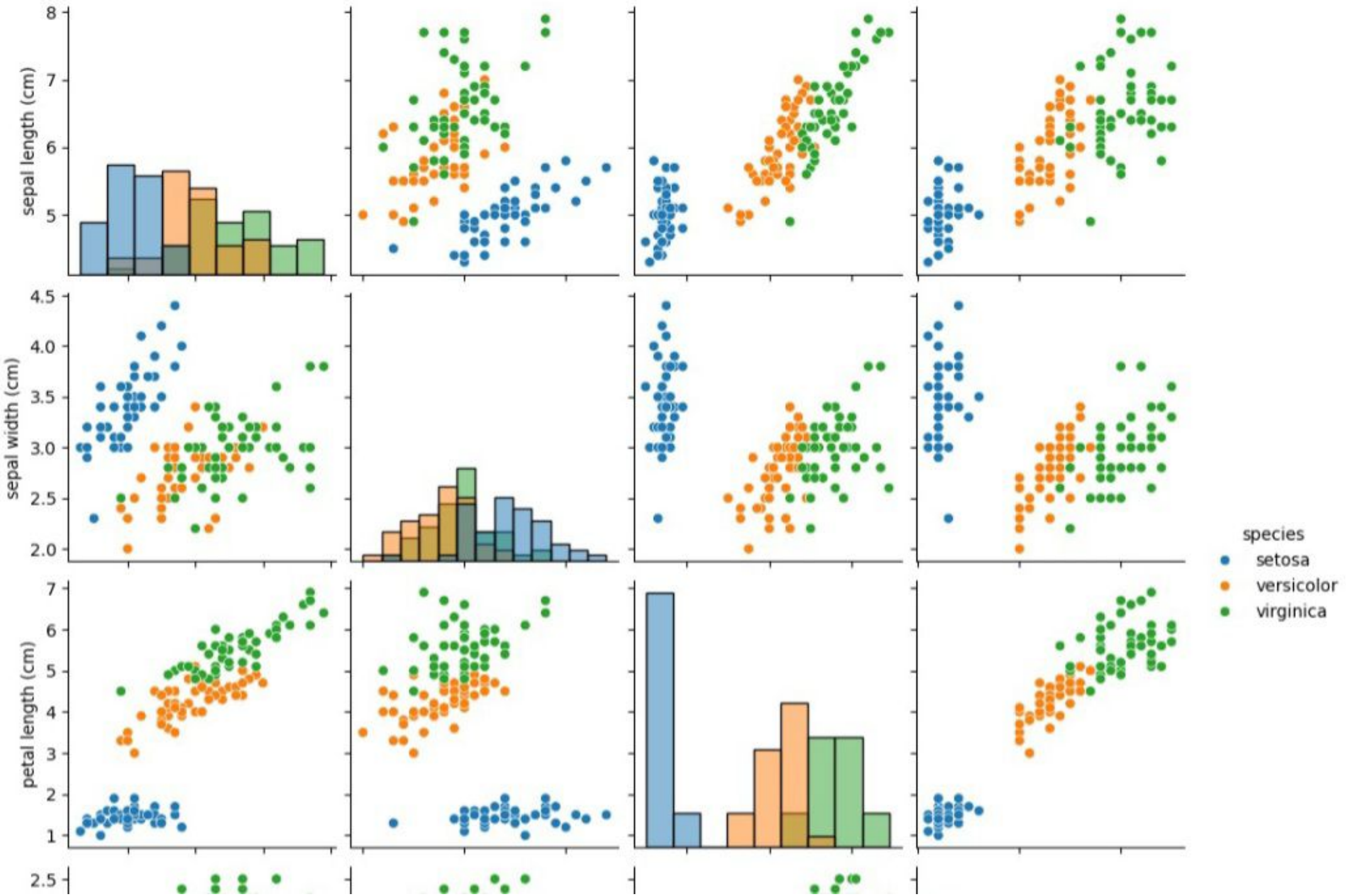
```
Dataset Shape: (150, 5)
Column Names:
Index(['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)',
       'petal width (cm)', 'species'],
      dtype='object')

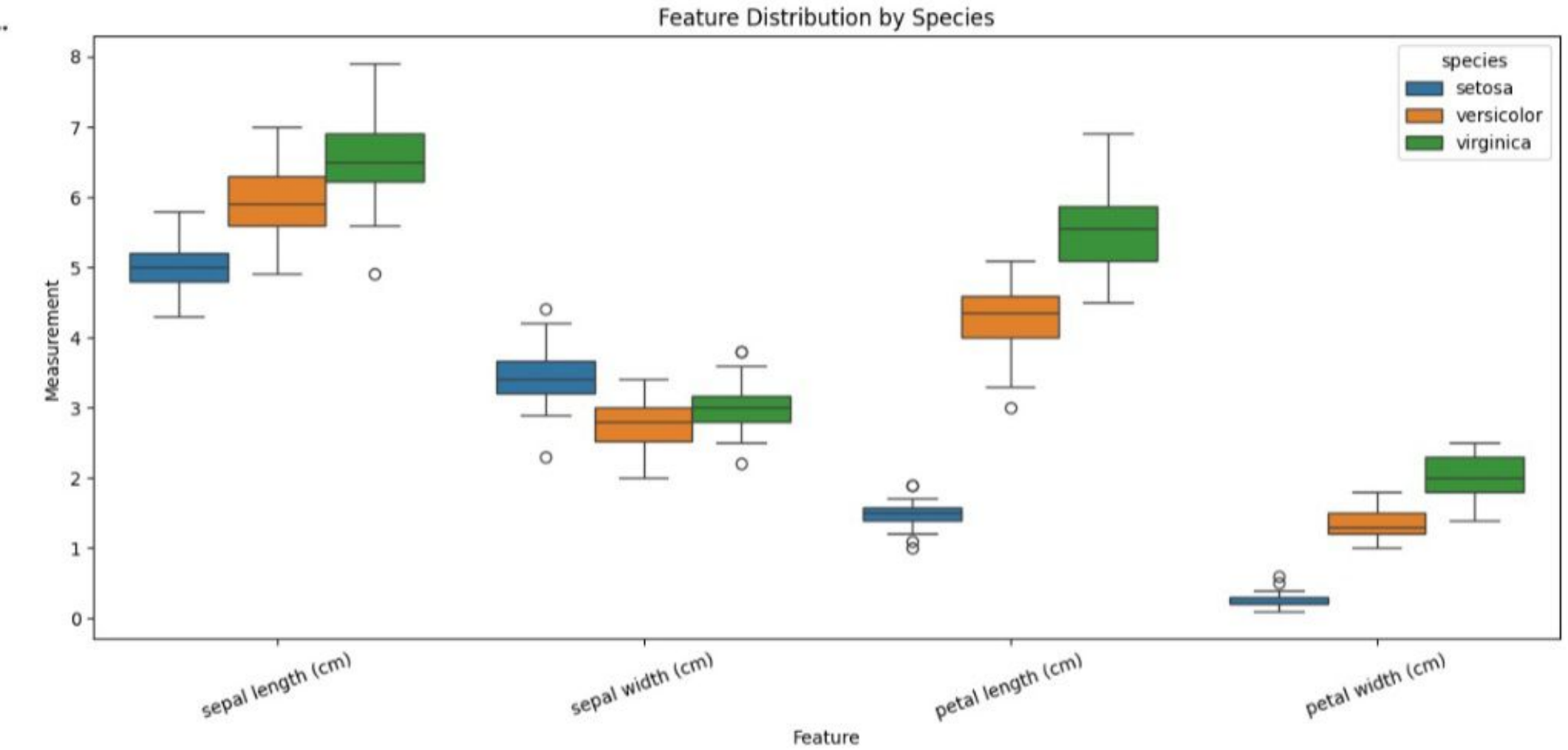
Data Types:
sepal length (cm)    float64
sepal width (cm)     float64
petal length (cm)    float64
petal width (cm)     float64
species               object
dtype: object
Missing Values:
sepal length (cm)    0
sepal width (cm)     0
petal length (cm)    0
petal width (cm)     0
species              0
dtype: int64
Descriptive Statistics:
```

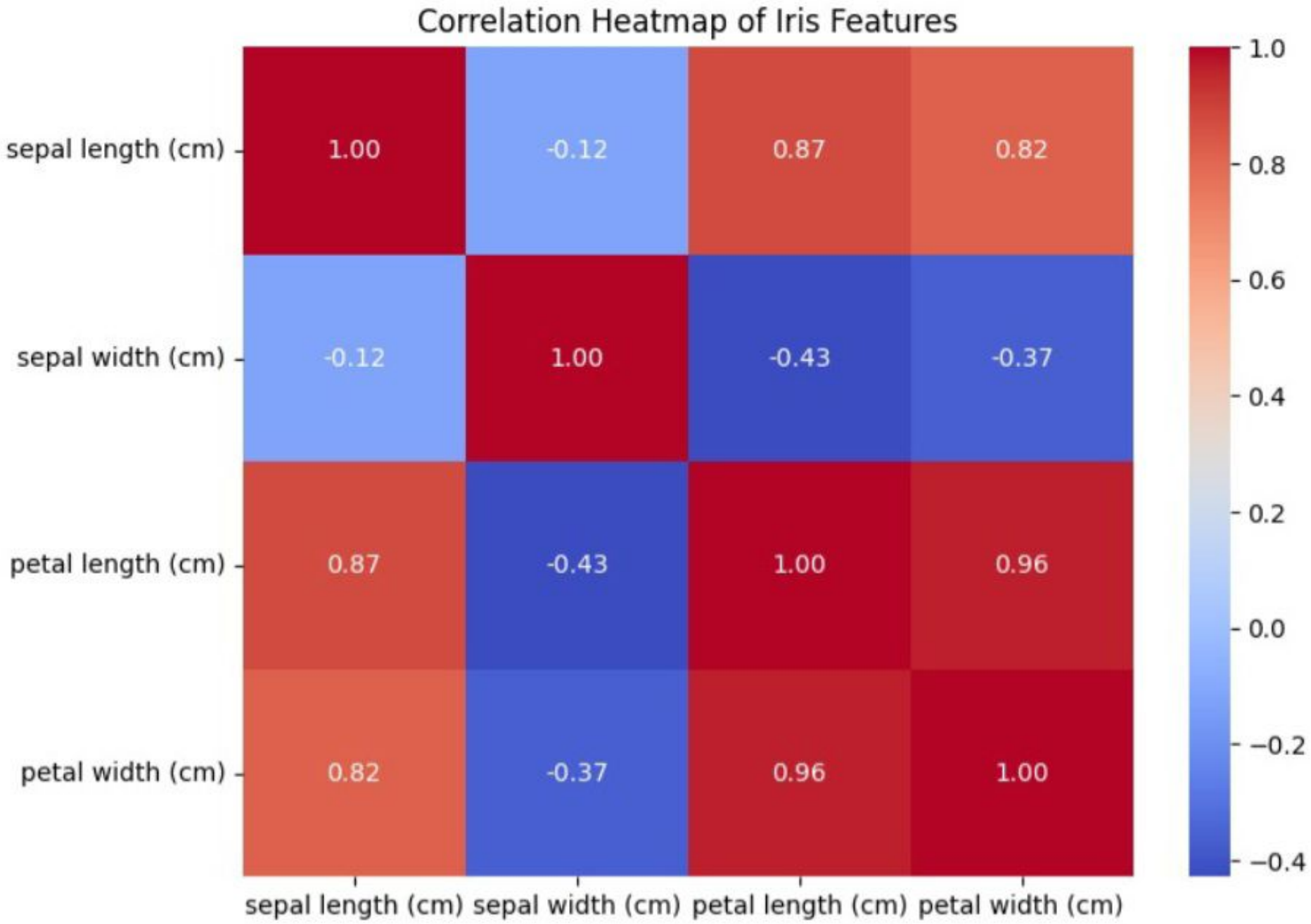
	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.199333
std	0.828066	0.435866	1.765298	0.762238
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

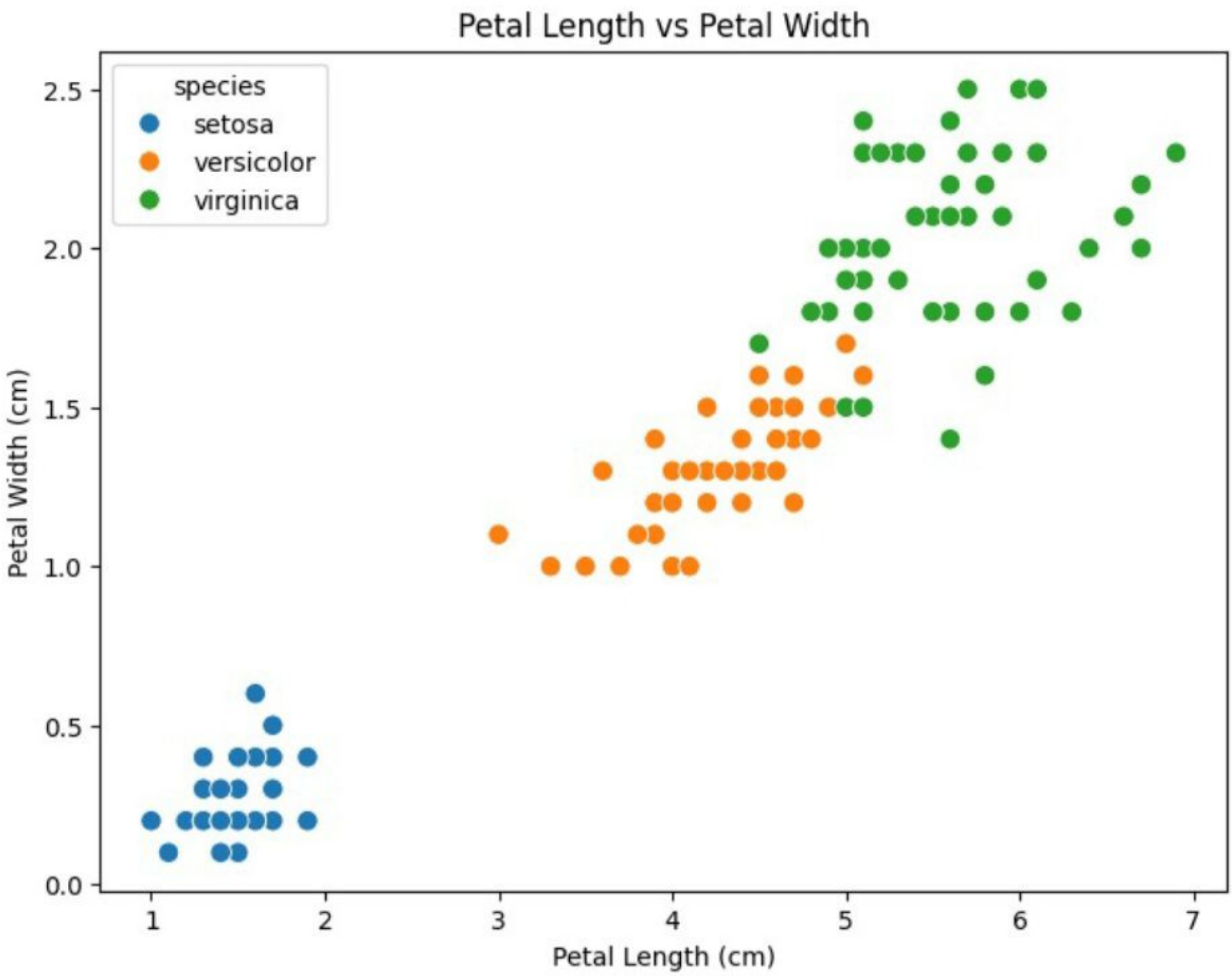
```
Species Distribution:
species
setosa        50
versicolor    50
virginica     50
Name: count, dtype: int64
```


Distribution of Iris Species
Count
50
40
30
20
10
0
setosa
versicolor
virginica
Species


sepal length (cm)
sepal width (cm)
petal length (cm)
8
7
6
5
4.5
4.0
3.5
3.0
2.5
2.0
7
6
5
4
3
2
1
2.5
species
setosa
versicolor
virginica

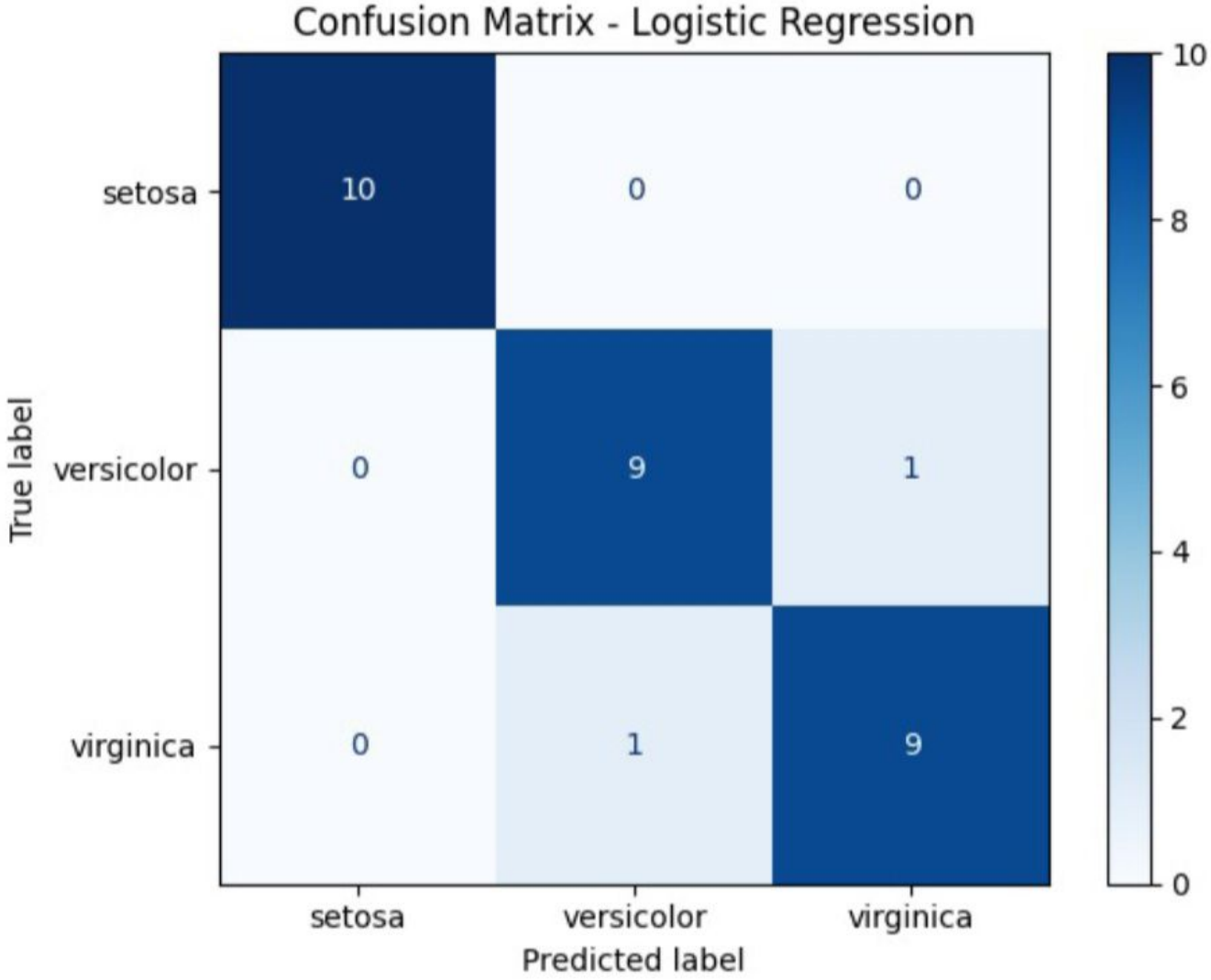

Feature Distribution by Species
species
setosa
versicolor
virginica
Measurement
0
1
2
3
4
5
6
7
8
sepal length (cm)
sepal width (cm)
petal length (cm)
petal width (cm)
Feature


Correlation Heatmap of Iris Features
sepal length (cm)
sepal width (cm)
petal length (cm)
petal width (cm)
1.00
-0.12
0.87
0.82
-0.12
1.00
-0.43
-0.37
0.87
-0.43
1.00
0.96
0.82
-0.37
0.96
1.00
sepal length (cm)
sepal width (cm)
petal length (cm)
petal width (cm)
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4


Petal Length vs Petal Width
species
setosa
versicolor
virginica
Petal Width (cm)
Petal Length (cm)
0.0
0.5
1.0
1.5
2.0
2.5
1
2
3
4
5
6
7

```
Logistic Regression trained successfully!
K-Nearest Neighbours trained successfully!
```

```
    accuracy                           0.93        30
   macro avg       0.93      0.93      0.93        30
weighted avg       0.93      0.93      0.93        30
```




```
==========================================
Model: K-Nearest Neighbours
==========================================
Accuracy: 0.9333

Classification Report:
              precision    recall  f1-score   support

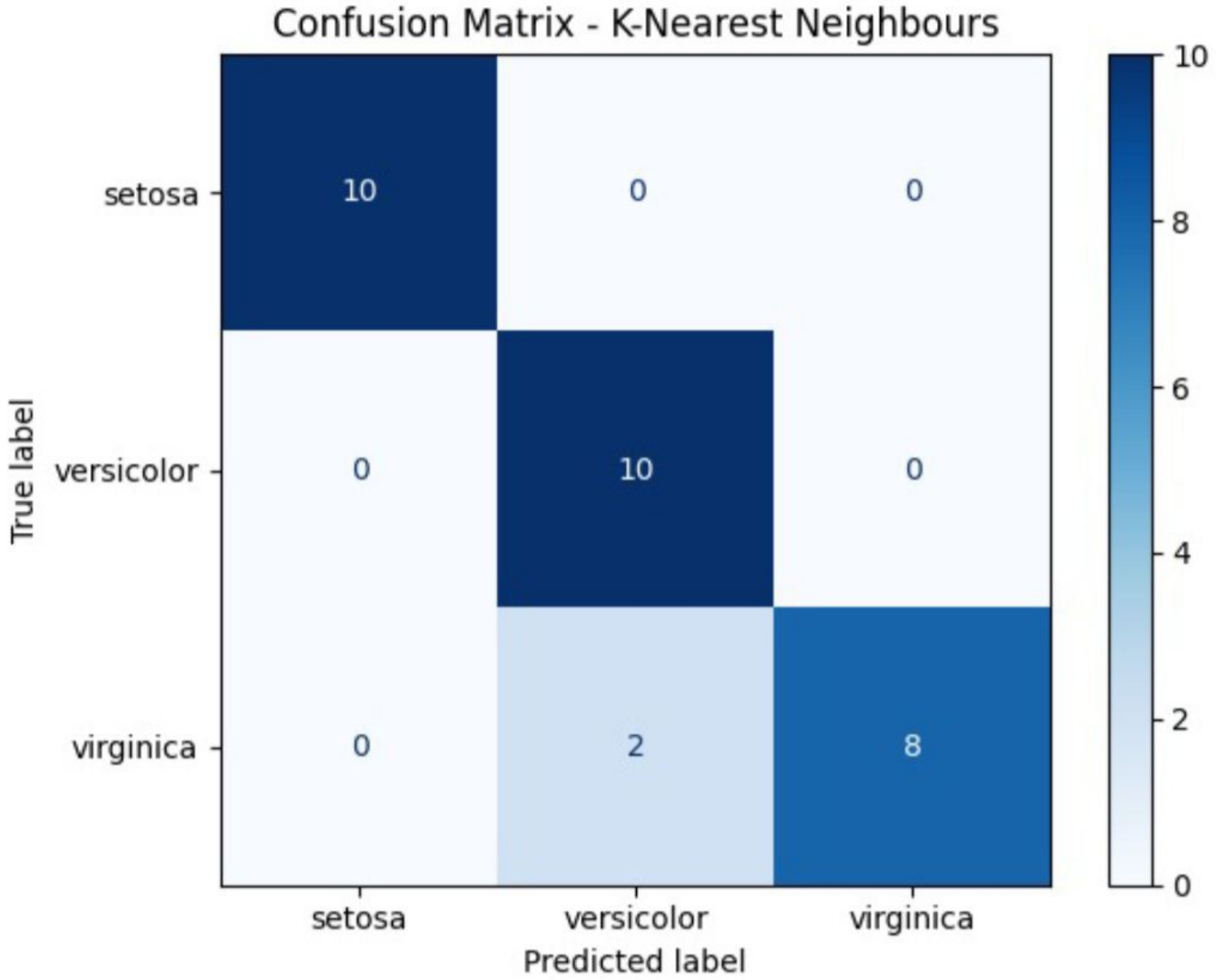
      setosa       1.00      1.00      1.00        10
  versicolor       0.83      1.00      0.91        10
   virginica       1.00      0.80      0.89        10
```

```
Model: K-Nearest Neighbours
==================================================
Accuracy: 0.9333

Classification Report:
              precision    recall  f1-score   support

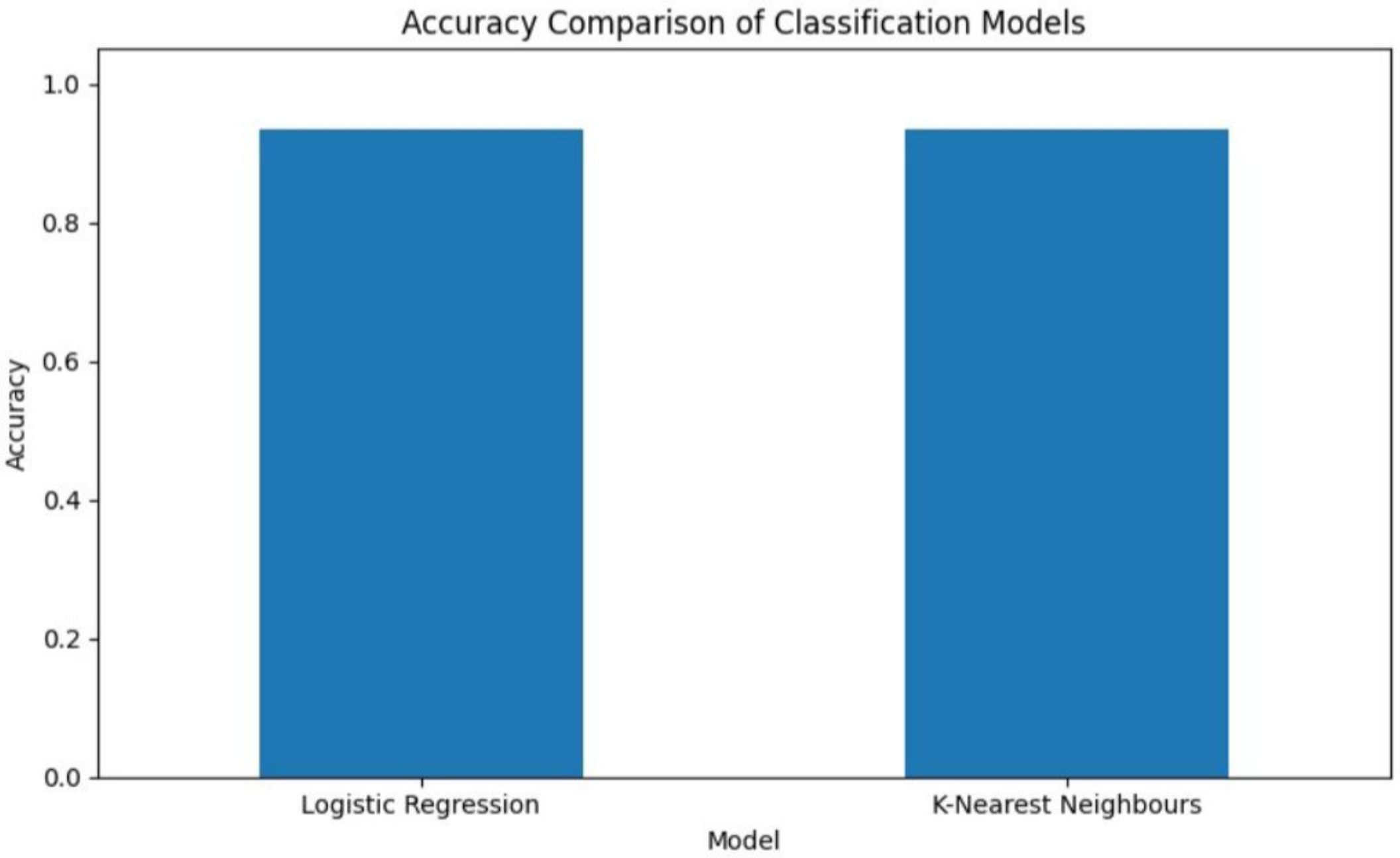
      setosa       1.00      1.00      1.00        10
  versicolor       0.83      1.00      0.91        10
   virginica       1.00      0.80      0.89        10

    accuracy                           0.93        30
   macro avg       0.94      0.93      0.93        30
weighted avg       0.94      0.93      0.93        30
```



```
Model Performance Comparison:
```

	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.9333	0.9333	0.9333	0.9333
K-Nearest Neighbours	0.9333	0.9444	0.9333	0.9327


Accuracy Comparison of Classification Models
1.0
0.8
0.6
0.4
0.2
0.0
Accuracy
Logistic Regression
K-Nearest Neighbours
Model

```
Best-Performing Model: Logistic Regression
Best Model Accuracy: 93.33%
Best Model F1-Score: 0.9333
```

```
Predicted Iris Species: setosa
```

```
================================================
IRIS FLOWER CLASSIFICATION - FINAL RESULTS
================================================

Dataset Information:
Total Samples: 150
Number of Features: 4
Number of Species: 3

Models Evaluated:
- Logistic Regression
- K-Nearest Neighbours

Model Comparison:
```

	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.9333	0.9333	0.9333	0.9333
K-Nearest Neighbours	0.9333	0.9444	0.9333	0.9327

```
Best-Performing Model: Logistic Regression
Test Accuracy: 93.33%

Project Completed Successfully!
```